

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	ATHIN SIVA.S	Reg. No.:	192524237
Section / Batch:	2023	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 –	Level 3 –	Level 2 –	Level 1 –
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		Exemplary	Proficient	Developing	Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO3 AT2-CASE STUDY

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Language: **Python** C C++ Java

QUESTIONS

q1 
Web Server Log Analysis (Sorting + Binary Search) A web server processes large log files that must be sorted and searched efficiently
100 marks

1/1 answered

q1 Web Server Log Analysis (Sorting + Binary Search) A web server processes large log files that must be sorted and searched efficiently

100 marks

Analyze how sorting algorithms and Binary Search can be combined. Identify issues such as processing time and large data volume. Apply divide and conquer techniques. Analyze overall complexity. Design a solution and justify its efficiency.

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 // Structure to represent a web server log entry
6 typedef struct {
7     long long timestamp; // Unix timestamp for sorting/searching
8     char ip_address[16];
9     char request[32];
10    int status_code;
11 } LogEntry;
12
13 // Merge function for Merge Sort (Divide and Conquer)
14 void merge(LogEntry arr[], int left, int mid, int right) {
15     int n1 = mid - left + 1;
16     int n2 = right - mid + 1;
```

Input

stdin input

Output